Institut National des Postes et Télécommunications
Master Data & Intelligence Artificielle

Devoir 1

Module : Fondements Mathématiques pour l'IA

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Interpolation Polynomiale – Interpolation de Lagrange

Question 1

La formule générale du polynôme d'interpolation de Lagrange est

$$P_n(x) = \sum_{i=0}^{n} f(x_i) L_i(x),$$

où

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}.$$

Nous allons écrire les expressions de $P_1(x)$, $P_2(x)$, $P_3(x)$ et $P_4(x)$.

1. Pour $n = 1$

Pour deux points

$$(x_0, f(x_0)), \qquad (x_1, f(x_1)),$$

on a

$$P_1(x) = f(x_0) L_0(x) + f(x_1) L_1(x).$$

Les polynômes de base sont

$$L_0(x) = \frac{x - x_1}{x_0 - x_1},$$

et

$$L_1(x) = \frac{x - x_0}{x_1 - x_0}.$$

Donc,

$$\boxed{P_1(x) = f(x_0)\frac{x - x_1}{x_0 - x_1} + f(x_1)\frac{x - x_0}{x_1 - x_0}}$$

2. Pour $n = 2$

Pour trois points

$$(x_0, f(x_0)), \qquad (x_1, f(x_1)), \qquad (x_2, f(x_2)),$$

on a

$$P_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x).$$

Les polynômes de base sont

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)},$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)},$$

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}.$$

Ainsi,

$$\boxed{\begin{aligned} P_2(x) = f(x_0)&\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} \\ &+ f(x_1)\frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} \\ &+ f(x_2)\frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned}}$$

3. Pour $n = 3$

Pour quatre points

$$(x_0, f(x_0)), \qquad (x_1, f(x_1)), \qquad (x_2, f(x_2)), \qquad (x_3, f(x_3)),$$

on a

$$P_3(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) + f(x_3)L_3(x).$$

Les polynômes de base sont

$$L_0(x) = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)},$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)},$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)},$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}.$$

Ainsi,

$$\boxed{\begin{aligned} P_3(x) = f(x_0)&\frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \\ &+ f(x_1)\frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \\ &+ f(x_2)\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \\ &+ f(x_3)\frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}. \end{aligned}}$$

4. Pour $n = 4$

Pour cinq points

$$(x_0, f(x_0)), \qquad (x_1, f(x_1)), \qquad (x_2, f(x_2)), \qquad (x_3, f(x_3)), \qquad (x_4, f(x_4)),$$

on a

$$P_4(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) + f(x_3)L_3(x) + f(x_4)L_4(x).$$

Les polynômes de base sont

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)},$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)},$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)},$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)},$$

$$L_4(x) = \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)}.$$

Ainsi,

$$\boxed{\begin{aligned} P_4(x) = f(x_0)&\frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} \\ &+ f(x_1)\frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} \\ &+ f(x_2)\frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} \\ &+ f(x_3)\frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} \\ &+ f(x_4)\frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)}. \end{aligned}}$$

Question 2

We are given the following three points:

$$(1,3), \qquad (10,7), \qquad (8,5).$$

Therefore,

$$x_0 = 1, \qquad f(x_0) = f(1) = 3,$$

$$x_1 = 10, \qquad f(x_1) = f(10) = 7,$$

and

$$x_2 = 8, \qquad f(x_2) = f(8) = 5.$$

Calculation of $L_0(x)$

$$\begin{aligned} L_0(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \\ &= \frac{(x-10)(x-8)}{(1-10)(1-8)} \end{aligned}$$

Since

$$(1-10)(1-8) = 63,$$

we obtain

$$\boxed{L_0(x) = \frac{(x-10)(x-8)}{63}}.$$

Calculation of $L_1(x)$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$= \frac{(x - 1)(x - 8)}{(10 - 1)(10 - 8)}$$

Since

$$(10 - 1)(10 - 8) = 18,$$

we obtain

$$\boxed{L_1(x) = \frac{(x - 1)(x - 8)}{18}}.$$

Calculation of $L_2(x)$

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$= \frac{(x - 1)(x - 10)}{(8 - 1)(8 - 10)}$$

Since

$$(8 - 1)(8 - 10) = -14,$$

we obtain

$$\boxed{L_2(x) = -\frac{(x - 1)(x - 10)}{14}}.$$

Construction of $P_2(x)$

Using

$$P_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x),$$

we obtain

$$P_2(x) = f(1)\frac{(x - 10)(x - 8)}{63} + f(10)\frac{(x - 1)(x - 8)}{18} - f(8)\frac{(x - 1)(x - 10)}{14}.$$

Since

$$f(1) = 3, \qquad f(10) = 7, \qquad f(8) = 5,$$

we have

$$P_2(x) = 3\frac{(x - 10)(x - 8)}{63} + 7\frac{(x - 1)(x - 8)}{18} - 5\frac{(x - 1)(x - 10)}{14}.$$

Expanding,

$$(x-10)(x-8) = x^2 - 18x + 80,$$

$$(x-1)(x-8) = x^2 - 9x + 8,$$

and

$$(x-1)(x-10) = x^2 - 11x + 10.$$

Therefore,

$$P_2(x) = \frac{x^2 - 18x + 80}{21} + \frac{7(x^2 - 9x + 8)}{18} - \frac{5(x^2 - 11x + 10)}{14}.$$

After simplification,

$$\boxed{P_2(x) = \frac{5x^2 - 27x + 211}{63}}.$$

Verification

At the three interpolation points:

$$P_2(1) = \frac{5(1)^2 - 27(1) + 211}{63} = 3,$$

$$P_2(10) = \frac{5(10)^2 - 27(10) + 211}{63} = 7,$$

$$P_2(8) = \frac{5(8)^2 - 27(8) + 211}{63} = 5.$$

Hence,

$$\boxed{P_2(1) = 3, \qquad P_2(10) = 7, \qquad P_2(8) = 5}$$

which confirms that the polynomial passes through the three given points.

Python Implementation

```
import numpy as np

# Interpolation points
x_i = np.array([1.0, 10.0, 8.0])
y_i = np.array([3.0, 7.0, 5.0])

def lagrange(x, x_i, y_i):
    P = 0.0

    for i in range(len(x_i)):
```

```
        L_i = 1.0

        for j in range(len(x_i)):
            if i != j:
                L_i *= (x - x_i[j]) / (x_i[i] - x_i[j])

        P += y_i[i] * L_i

    return P


# Check the interpolation
for x, y in zip(x_i, y_i):
    print(f"P({x}) = {lagrange(x, x_i, y_i)}")
    print(f"Expected = {y}")
    print()

# Interpolation at x = 5
x = 5

print(f"P({x}) = {lagrange(x, x_i, y_i)}")
```

Python Result

```
Checking the interpolation at the given points:
P(1.0) = 3.0
Expected value = 3.0

P(10.0) = 7.0
Expected value = 7.0

P(8.0) = 5.0
Expected value = 5.0

Interpolation at x = 5:
P(5) = 3.1904761904761907
```